

ANDEER, Switzerland

WMO: 067870

Lat: 46.6101N

Lon: 9.4319E

Elev: 3242

StdP: 13.05

Time Zone: 1.00 (EUC)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 8.0	(c) 12.7	(d) 2.4	(e) 7.0	(f) 12.2	(g) 6.9	(h) 8.8	(i) 15.5	(j) 19.0	(k) 41.4	(l) 16.6	(m) 41.1	(n) 1.5	(o) 120	(p) 0.554

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 21.3	(c) 83.6	(d) 63.1	(e) 80.4	(f) 62.2	(g) 76.9	(h) 60.9	(i) 64.8	(j) 78.7	(k) 63.5	(l) 76.5	(m) 62.1	(n) 73.8	(o) 8.6	(p) 20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
60.6	89.0	68.1	59.1	84.5	66.6	57.8	80.5	65.7	31.7	79.2	30.5	77.0	29.5	73.7	71.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 15.9	(b) 13.5	(c) 11.6	(e) 3.6	(f) 89.0	(g) 5.8	(h) 1.3	(i) -0.6	(j) 89.9	(k) -4.1	(l) 90.6	(m) -7.3	(n) 91.3	(o) -11.6	(p) 92.3		
(a) 15.9	(b) 13.5	(c) 11.6	(e) 2.7	(f) 68.3	(g) 5.8	(h) 1.7	(i) -1.4	(j) 69.6	(k) -4.8	(l) 70.5	(m) -8.0	(n) 71.5	(o) -12.2	(p) 72.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	46.5	28.5	30.7	38.9	46.6	52.4	60.5	63.0	62.8	55.3	48.1	39.2	31.3		
	DBStd	13.74	7.01	8.35	6.14	6.19	6.10	5.90	5.45	5.36	5.36	6.73	7.35	7.31		
	HDD50	2767	666	539	345	134	46	2	0	0	15	116	325	579		
	HDD65	6840	1131	959	809	553	390	160	101	106	291	523	773	1044		
	CDD50	1504	0	0	1	31	121	315	404	397	174	58	3	0		
	CDD65	103	0	0	0	0	0	23	40	39	1	0	0	0		
	CDH74	1376	0	0	0	0	40	370	488	449	29	0	0	0		
	CDH80	309	0	0	0	0	3	86	116	104	0	0	0	0		
	WSAvg	3.2	2.4	2.9	3.6	4.3	4.2	3.7	3.4	3.0	2.6	2.9	2.7	2.5		
	PrecAvg	47.6	2.2	2.1	2.4	3.4	4.5	5.4	5.3	5.8	4.7	4.4	4.9	2.5		
(15) Precipitation	PrecMax	66.4	4.6	6.6	5.7	8.6	11.9	10.6	8.7	9.3	10.8	13.7	20.2	5.2		
	PrecMin	31.8	0.4	0.6	0.3	0.6	1.7	1.6	2.6	0.8	1.2	0.4	1.1	0.1		
	PrecStd	8.6	1.3	1.5	1.3	1.9	2.4	2.6	1.7	1.8	3.0	2.8	4.1	1.3		
	PrecStd	8.6	1.3	1.5	1.3	1.9	2.4	2.6	1.7	1.8	3.0	2.8	4.1	1.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.2	52.5	63.1	71.4	79.1	86.8	87.6	87.2	77.4	70.1	60.0	51.5		
		MCWB	39.7	40.1	45.6	52.5	58.7	63.5	64.5	63.5	59.8	54.6	47.9	42.3		
	2%	DB	44.7	47.4	58.3	66.8	73.6	82.4	83.2	82.8	73.8	66.1	56.0	48.3		
		MCWB	37.6	37.9	44.0	51.0	56.1	62.4	63.3	63.1	58.9	53.2	46.3	40.5		
	5%	DB	41.9	44.4	54.2	62.7	69.4	78.3	79.8	79.3	69.6	62.9	53.6	45.7		
		MCWB	36.0	36.6	42.6	48.5	54.4	61.0	62.2	62.3	57.1	51.5	45.1	38.6		
	10%	DB	38.6	41.7	50.1	59.4	65.4	74.1	75.7	75.2	65.8	59.4	51.0	42.5		
		MCWB	34.5	35.1	40.9	46.9	52.7	59.1	61.1	61.2	55.9	50.6	43.8	36.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	40.9	41.9	47.0	53.4	59.7	66.3	66.6	66.8	61.9	56.6	49.3	43.9		
		MCDB	46.2	49.3	59.3	68.2	75.4	82.7	81.2	78.4	72.4	66.3	56.6	48.7		
	2%	WB	38.7	38.9	44.9	51.4	57.5	63.8	64.8	64.7	60.1	54.6	47.2	41.5		
		MCDB	43.4	45.0	56.1	65.1	71.2	77.8	79.2	77.9	70.3	62.2	54.1	47.4		
	5%	WB	36.8	37.2	43.1	49.4	55.3	61.9	63.5	63.3	58.5	53.1	45.7	39.4		
		MCDB	40.6	43.2	52.4	60.6	66.6	75.6	76.5	75.7	66.7	59.6	52.1	44.4		
	10%	WB	34.9	35.7	41.3	47.5	53.5	60.2	62.0	62.0	56.9	51.6	44.3	37.2		
		MCDB	37.9	41.0	48.9	57.6	63.7	71.7	73.1	72.9	63.8	58.0	50.2	41.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	13.9	16.0	18.6	20.1	19.5	21.0	21.3	21.4	19.3	18.2	14.7	13.5		
		MCDBR	17.0	19.8	26.4	27.0	27.8	29.6	29.0	28.7	26.3	24.6	19.5	17.8		
		MCWBR	11.0	11.9	14.2	13.5	12.9	12.3	11.9	12.0	13.0	13.1	10.9	10.8		
		MCDBR	14.4	18.3	23.5	25.1	24.9	26.2	25.0	24.9	22.2	20.4	17.9	15.3		
	5% WB	MCWBR	9.7	11.3	12.9	12.9	11.9	11.4	11.4	11.5	11.4	11.7	10.3	9.7		
(40) Clear-Sky Solar Irradiance	taub		0.256	0.283	0.330	0.381	0.397	0.407	0.402	0.394	0.366	0.335	0.289	0.253		
	taud		2.316	2.242	2.172	2.146	2.230	2.257	2.276	2.318	2.374	2.433	2.432	2.356		
	Ebn at Noon		269	282	282	276	275	272	272	270	268	260	254	257		
	Edh at Noon		23	31	40	45	43	42	41	37	32	26	21	20		
(44) All-Sky Solar Radiation	RadAvg		496	800	1184	1413	1520	1785	1777	1543	1198	793	487	413		
	RadStd		61	94	170	227	153	160	155	139	118	93	72	54		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (117 neighbors)	N/A	N/A	N/A	+1.26	+0.72	+0.58	N/A	N/A	+106	+29

Nomenclature: See separate page